MOTOR (Steven)

Rpm

→ C++

→ Arduino Library

→ The computer is wobbling on one foot trying to keep itself as balanced as possible, if it overcorrects it can reconnect

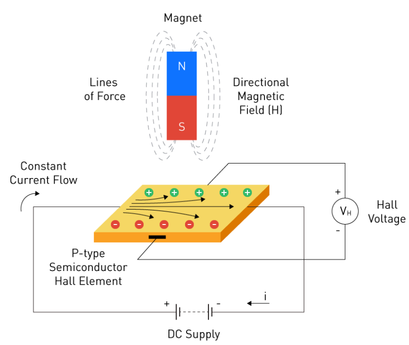
Main.cpp

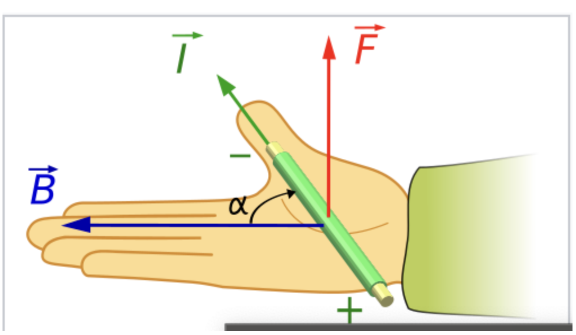
Hall effect sensor

→ A Hall effect sensor is a transducer (a device that converts energy from one form to another) that varies its output voltage in response to an external magnetic field, acting as a non-contact, durable, and highly accurate magnetic sensor. Used for position detection, speed measurement, and proximity switching, these devices operate by detecting the Lorentz force on charge carriers in a semiconductor, producing a measurable

→ The Lorentz force is the total electromagnetic force exerted on a charged particle moving with velocity through both an electric field and a magnetic field.

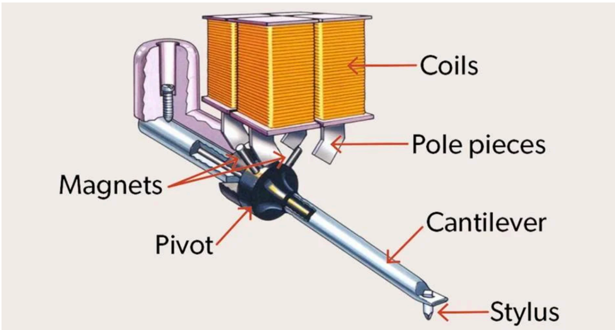
$$\vec{F} = \underbrace{q\vec{E}}_{\text{Electric force}} + \underbrace{q\vec{v} \times \vec{B}}_{\text{Magnetic force}}$$

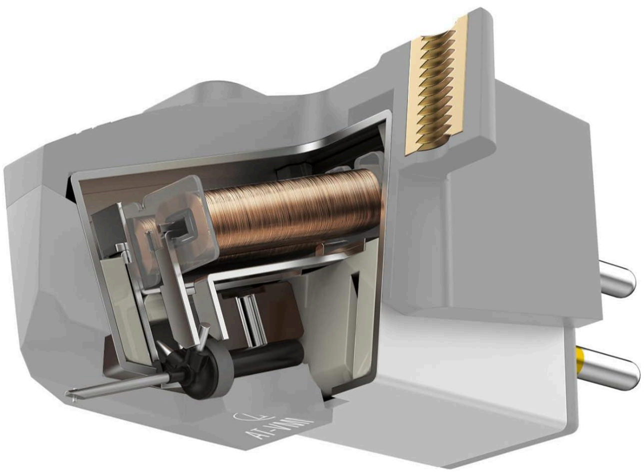




→ Electric force is parallel to the electric field. Magnetic force is perpendicular to the velocity and magnetic field.

TONE ARM





<u>The vibrations are converted into electrical signals</u>

The stylus connected to the cantilever that interacts with the disc
Magnets connected to the pivot

Faraday's law
<u>Mechanical to electrical</u>